WORKED SOLUTION

STEP III 2002, Question 2

Pure • Inverse Tangent Series

STEP III 2002, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q2

The key idea

The first result is the tangent addition formula in disguise. The only real work is checking that the two sides are angles in the same interval, where tan is one-to-one. Once we have that formula, the induction is one application of it plus some factorising. For the last sum, the trick is to notice that $r^2 - r + 1$ is just $r^2 + r + 1$ with r shifted by one. So the new series is the old one with a single extra term at the front.

Solution

The addition formula

Let $A = \arctan a$ and $B = \arctan b$. Then $\tan A = a$ and $\tan B = b$, and

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} = \frac{a + b}{1 - ab},$$

which makes sense because $ab < 1$. This shows that $A + B$ and $\arctan(\frac{a+b}{1-ab})$ have the same tangent. That alone doesn't make them equal, since tan repeats every π. So we need both angles in one interval on which tan is one-to-one.

? **Ask yourself** Why isn't $\tan(A + B) = \frac{a+b}{1-ab}$ already the end of the proof?

It isn't the end because arctan only returns values in $(-\frac{\pi}{2}, \frac{\pi}{2})$. We must check that $A + B$ lies there too.

- Since $0 < a < 1$ and arctan is increasing, $0 < A < \frac{\pi}{4}$. Similarly $0 < B < \frac{\pi}{4}$, so $0 < A + B < \frac{\pi}{2}$.
- Since $a + b > 0$ and $1 - ab > 0$, the fraction $\frac{a+b}{1-ab}$ is positive. So its arctan also lies in $(0, \frac{\pi}{2})$.

The function tan is strictly increasing, and hence one-to-one, on $(0, \frac{\pi}{2})$. Two angles

there with the same tangent are therefore equal, which gives

$$\arctan a + \arctan b = \arctan\left(\frac{a+b}{1-ab}\right). \qquad \checkmark$$

The induction

Write $S_n = \sum_{r=1}^{n} \arctan\left(\frac{1}{r^2+r+1}\right)$. The claim is $S_n = \arctan\left(\frac{n}{n+2}\right)$.

Base case. When $n = 1$, both sides are $\arctan\frac{1}{3}$.

Inductive step. Suppose $S_n = \arctan\left(\frac{n}{n+2}\right)$ for some $n \geq 1$. The next term has

$$(n+1)^2 + (n+1) + 1 = n^2 + 3n + 3,$$

so

$$S_{n+1} = \arctan\left(\frac{n}{n+2}\right) + \arctan\left(\frac{1}{n^2+3n+3}\right).$$

Both arguments lie strictly between 0 and 1, so the addition formula applies with $a = \frac{n}{n+2}$ and $b = \frac{1}{n^2+3n+3}$. Multiplying the top and bottom of $\frac{a+b}{1-ab}$ by $(n+2)(n^2+3n+3)$ gives

$$\begin{aligned}\frac{a+b}{1-ab} &= \frac{n(n^2+3n+3)+(n+2)}{(n+2)(n^2+3n+3)-n} \\ &= \frac{n^3+3n^2+4n+2}{n^3+5n^2+8n+6}.\end{aligned}$$

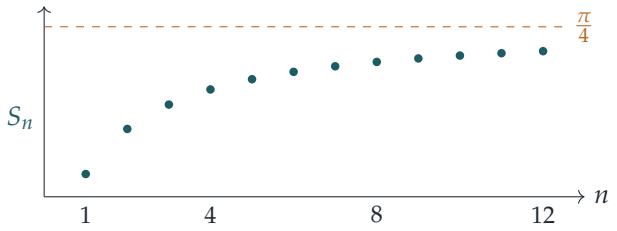
Both cubics vanish at a negative integer: the top at $n = -1$ and the bottom at $n = -3$. Factorising,

$$\frac{a+b}{1-ab} = \frac{(n+1)(n^2+2n+2)}{(n+3)(n^2+2n+2)} = \frac{n+1}{n+3},$$

where we can cancel because $n^2+2n+2 = (n+1)^2+1 > 0$. Hence $S_{n+1} = \arctan\left(\frac{n+1}{(n+1)+2}\right)$, which is the statement for $n+1$. By induction, the result holds for all $n \geq 1$. ✓

The infinite sum. As $n \to \infty$ we have $\frac{n}{n+2} \to 1$. Since arctan is continuous,

$$\sum_{r=1}^{\infty} \arctan\left(\frac{1}{r^2+r+1}\right) = \lim_{n\to\infty} \arctan\left(\frac{n}{n+2}\right) = \arctan 1 = \frac{\pi}{4}.$$



The final sum

Put $s = r - 1$. Then

$$r^2 - r + 1 = (s+1)^2 - (s+1) + 1 = s^2 + s + 1.$$

As r runs over $1, 2, 3, \ldots$, s runs over $0, 1, 2, \ldots$. The $s = 0$ term is $\arctan 1 = \frac{\pi}{4}$, and the remaining terms ($s \geq 1$) make up exactly the series we have just summed. Therefore

$$\sum_{r=1}^{\infty} \arctan\left(\frac{1}{r^2 - r + 1}\right) = \arctan 1 + \sum_{s=1}^{\infty} \arctan\left(\frac{1}{s^2 + s + 1}\right) = \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}. \qquad \checkmark$$

The second series converges because the first one does, so splitting off the first term is legitimate.

Checks

- When $n = 2$: $\arctan\frac{1}{3} + \arctan\frac{1}{7}$ has tangent $\frac{10/21}{20/21} = \frac{1}{2} = \frac{2}{4}$, as the formula predicts.
- Numerically, $S_{2000} \approx 0.784898$, while $\arctan\frac{2000}{2002}$ agrees to 10^{-12} and $\frac{\pi}{4} \approx 0.785398$.
- Summing the second series to $r = 10^6$ gives $1.5707953\ldots$, which is within 10^{-6} of $\frac{\pi}{2}$. That matches the gap after n terms, $\frac{\pi}{4} - \arctan\frac{n-1}{n+1} = \arctan\frac{1}{n} \approx \frac{1}{n}$.

Common slips

- Stopping at "tan of both sides is equal". Without the range argument, the proof is incomplete. (For example, with $a = b = 2$ the formula fails: the left side is positive but $\arctan(-\frac{4}{3})$ is negative.)
- Forgetting to check that both arguments in the inductive step lie in $(0, 1)$ before using the formula.
- Writing the final sum as $\frac{\pi}{4}$ again. The shift $r \mapsto r - 1$ adds the extra term $\arctan 1$ at the front.

Beyond STEP

There is a slicker route to the induction. Since $\frac{1}{r^2+r+1} = \frac{(r+1)-r}{1+r(r+1)}$, each term is $\arctan(r+1) - \arctan r$. The sum therefore *telescopes* to $\arctan(n+1) - \frac{\pi}{4}$, which is $\arctan\frac{n}{n+2}$ by the subtraction formula. Spotting a telescoping structure and justifying limits of partial sums by continuity are everyday tools in a first analysis course. Abbott's *Understanding Analysis* treats series and continuity carefully, and Siklos's *Advanced Problems in Mathematics* has more STEP-style practice of this kind.

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